1) a. expected value of single dice roll

x	P
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

$\rightarrow 1\frac{1}{6} + 2\frac{1}{6} + 3\frac{1}{6} \dots + 6\frac{1}{6} = \frac{21}{6}$

b. expected value of rolling 2 dice & adding results together?

x	P
2	1/36
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	

$\rightarrow 2\frac{1}{36} + 3\frac{1}{36} + \dots 12\frac{1}{36} = \frac{252}{36}$

c. expected winnings of euro roulette

$\frac{36}{K}$ } gain; where K is # of slots you bet on

$\frac{K}{37}$ } probability of win

$\frac{37-K}{37}$ } prob. of losing

0 } gain when lose

$\mathbb{E}[x] = \frac{36}{\cancel{K}} \cdot \frac{\cancel{K}}{37} + \frac{37-K}{37} \cdot 0 - 1$

$\mathbb{E}[x] = \frac{36}{37} + -1 \rightarrow \frac{36}{37} - \frac{37}{37} = \frac{-1}{37}$

d. roll die & record value; if roll 6, roll again & add value; expected value?

$\mathbb{E}[x] = \frac{1}{6} + 2\frac{1}{6} + 3\frac{1}{6} + 4\frac{1}{6} + 5\frac{1}{6} + \frac{1}{6}\left[6 + \frac{1}{6} + 2\frac{1}{6} + 3\frac{1}{6} + \dots 6\frac{1}{6}\right]$

$\hookrightarrow \underbrace{\dots\dots\dots + \frac{1}{6}[6 +}_{\frac{21}{6}} \frac{21}{6}] \rightarrow \underset{\downarrow \frac{126}{6}}{\frac{21}{6}} + \frac{21}{36} = \frac{147}{36} = 4.083$

e. process from d until fail to roll 6; expected value?

$$R = \frac{1}{6} + 2\frac{1}{6} + 3\frac{1}{6} + 4\frac{1}{6} + 5\frac{1}{6} + \frac{1}{6}[6 + R]$$

$$R = \frac{21}{6} + \frac{R}{6}$$

$$R = \frac{21}{5}$$

2) a. expected value of uniform random variable

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x)\,dx \rightarrow f(x) = \begin{cases} \frac{1}{d-c} & c \le x \le d \\ 0 & \end{cases}$$

$$\mathbb{E}[X] = \int_c^d x \cdot \frac{1}{d-c}\,dx$$

$$\hookrightarrow \frac{1}{d-c}\int_c^d x \cdot dx = \frac{1}{d-c}\left[\frac{x^2}{2}\right]_c^d = \frac{1}{d-c}\left(\frac{d^2}{2} - \frac{c^2}{2}\right)$$

$$= \frac{1}{1-0}\left(\frac{1^2}{2} - \frac{0^2}{2}\right) = \frac{1}{2}$$

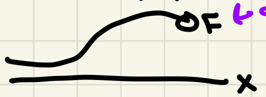
b. show that $\mathbb{E}[a+bX] = a + b\mathbb{E}(X)$

$$\mathbb{E}[a+bX] = \int (a+bx) f(x)\,dx$$

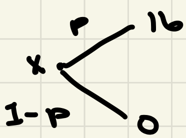
$$\int_{-a}^{x} f(x)\,dx = F(x) \qquad = \underbrace{\int a f(x)\,dx}_{\hookrightarrow = 1} + \underbrace{b\int x f(x)\,dx}_{\mathbb{E}(X)} = a + b\mathbb{E}[X]$$

$$\lim_{x \to a} \int_{-\infty} f(x)\,dx$$

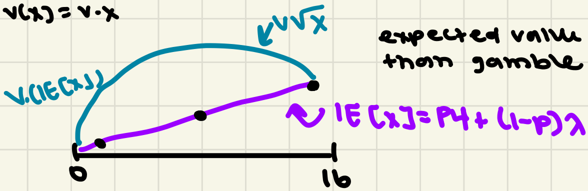
$$= \lim_{x \to \infty} F(x) = 1$$



c. $v(\mathbb{E}[X]) \neq \mathbb{E}[v(X)]$ if $v(x) \neq a + bx$



$v(x) = v \cdot x$



expected value higher than gamble value

3) a. variance for uniform random variable

$$\mathbb{V}[X] = \frac{(a-b)^2}{12} \to \frac{(1-0)^2}{12} = \frac{1}{12}$$

b. $\mathbb{V}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$

$\hookrightarrow \mathbb{V}[X] = \int_x (x - \mathbb{E}[X])^2 f(x)dx$

$\hookrightarrow \int_x (x^2 - 2x\mathbb{E}[X] - \mathbb{E}[X]^2) f(x)dx$

$\hookrightarrow \int_x x^2 f(x)dx - 2\int_x x f(x)dx\, \mathbb{E}[X] + \mathbb{E}[X]^2$

$\hookrightarrow \mathbb{E}[X^2]$ $\quad$ 2 $\quad$ $\hookrightarrow \mathbb{E}[X] \cdot \mathbb{E}[X]$

$\hookrightarrow \mathbb{E}[X^2] - 2(\mathbb{E}[X])^2 + \mathbb{E}[X]^2$

$\hookrightarrow -2+1 = -1$

$\hookrightarrow \mathbb{E}[X^2] - \mathbb{E}[X]^2$

$\mathbb{V}[a+bX] = b^2\mathbb{V}[X]$

$\hookrightarrow \mathbb{V}[a+bX] = \int (a+bX - \mathbb{E}[a+bX])^2 f(x)dx$

$\hookrightarrow \int (\cancel{a} + bX - \cancel{a} - b\mathbb{E}[X])^2 f(x)d(x)$

$\hookrightarrow \int (bX - b\mathbb{E}[X])^2 f(x)dx$

$\hookrightarrow b^2 \int (X - \mathbb{E}[X])^2 f(x)dx$

$\hookrightarrow = \mathbb{V}[X]$

$\hookrightarrow b^2\mathbb{V}[X]$

c. if X normally distr., then $a+bX$ distr. normally w/ mean $a + b\mathbb{E}[X]$ & variance $b^2\sigma^2_X$

$Y = a+bX$

$\hookrightarrow \frac{Y-a}{b} = X$

$$F_X\left(\frac{Y-a}{b}\right) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left(\frac{(Y-a)/b - \mu}{\sigma}\right)^2\right\}dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left(\frac{Y-\mu'}{\sigma'}\right)^2\right\}dx$$

5. a. expectation of ε

$$\boxed{\varepsilon = X - \mathbb{E}[X]}$$

$$\mathbb{E}[\varepsilon] = \int [x - \mathbb{E}[x]] f(x)\,dx$$

$$= \int x f(x)\,dx - \mathbb{E}[x]$$

$$= \mathbb{E}[x] - \mathbb{E}[x] = 0$$

b. variance of ε

$$\mathbb{V}[\varepsilon] = \int (\varepsilon - \underbrace{\mathbb{E}[\varepsilon]}_{=\emptyset \Rightarrow \text{cancels}})^2 f(\varepsilon)\,d\varepsilon$$

$$= \int (x - \mathbb{E}[x])^2 f(x)\,dx$$

$$= \mathbb{V}[x]$$

c. random variable in form $X = \mathbb{E}[X] + \sigma_x \varepsilon$, where $\mathbb{E}[\varepsilon] = \emptyset$ & $V[\varepsilon] = 1$

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[x]] + \mathbb{E}[\sigma_x \varepsilon]$$

$$= \mathbb{E}[x] + \sigma_x \cdot \emptyset$$

$$= \mathbb{E}[X]$$

$$\mathbb{V}[X] = \int (\mathbb{E}[x] + \sigma_x \varepsilon - \mathbb{E}[\mathbb{E}[x] + \sigma_x \varepsilon])^2 f(x)\,dx$$

$$= \int (\mathbb{E}[x] + \sigma_x \varepsilon - \mathbb{E}[x] - \sigma_x \mathbb{E}[\varepsilon])^2 f(x)\,dx$$

$$= \int (\sigma_x \varepsilon - \sigma_x \mathbb{E}[\varepsilon])^2 f(x)\,dx$$

$$= \int \sigma_x^2 (\varepsilon - \mathbb{E}[\varepsilon])^2 f(x)\,dx$$

$$= \sigma_x^2 \underbrace{\mathbb{V}[\varepsilon]}_{=1}$$

$$= \sigma_x^2$$

6. $\mathbb{E}[\hat{f}_{X,h}(x)] = \dfrac{F(x+h) - F(x-h)}{2h}$

$\hookrightarrow \dfrac{F(x) + hf(x) + \frac{h^2}{2}f'(x) + O(h^3) - F(x) + hf(x) - \frac{h^2}{h}f'(x) - O(h^3)}{2h}$

$\hookrightarrow \dfrac{2hf(x) + O(h^3) + hf(x) - O(h^3)}{2h}$

$\hookrightarrow f(x) + \dfrac{O(h^3) - O(h^3)}{2h}$

$\hookrightarrow f(x) + O(h^2)$

bandwidth small, then KDE = $f(x)$ + error on order of h^2

$\hookrightarrow$ if h small, error pretty small